

Introduction about Datacenters

Respondent: 265149-周明宇

Course: Computer Networking

2026/10/26

Contents

01 What is a datacenter

02 Types of datacenters

03 Components of datacenter

04 Serving requests

05 Reliability& Performance

06 Trends & summary

What is a datacenter?

SOUTHEAST UNIVERSITY 東南大學

Definition

A dedicated facility that houses many servers, storage and network gear, with the power, cooling and security to run them 24/7. The servers are end systems at the network edge; the core only forwards packets. Large providers (Google, Microsoft) link their datacenters with private networks to bring content close to users. A big site can cover tens of thousands of m^2 and draw tens to hundreds of MW — about a small city.

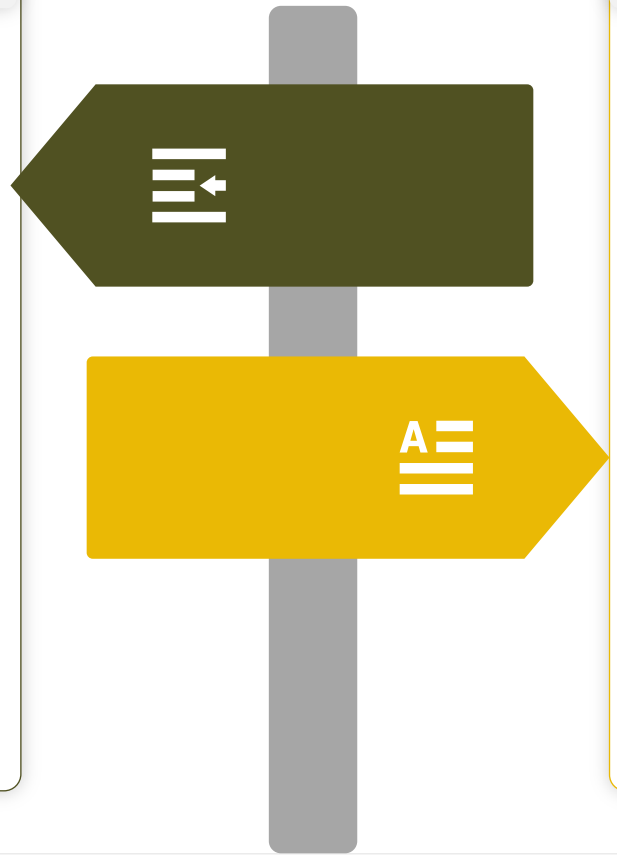
Types of datacenters

Type	Operated by	Typical scale	Example
Enterprise	A single organization for its own IT systems	Server room to a few thousand servers	A bank's private datacenter
Colocation	A provider renting space, power and network to many tenants	Shared building, many customers	Equinix, Digital Realty
Hyperscale	Cloud and Internet companies serving millions of users	Often 100,000+ servers per site	AWS, Google, Microsoft, Alibaba Cloud
Edge	Operators placing small sites close to end users	A few racks to a small building	CDN nodes, 5G edge computing

Components: IT equipment and facility

IT equipment

- Servers in 19-inch racks run apps and virtual machines.
- Storage arrays hold user data, replicated for safety.
- Switches/routers link servers at 10-100 Gbps and reach ISPs.
- Virtualization and cluster managers place and watch workloads.



Facility infrastructure

- Power: utility feed, UPS and generators for grid failures.
- Cooling: air or liquid removes the heat from all that power.
- Security: access control, cameras, clean fire suppression.
- Operations: monitoring and 24/7 staff fix faults fast.

How a datacenter serves user request

Every step is replicated across many machines and often several datacenters, so the whole trip should take well under a second.

1. DNS resolves the hostname, often to a nearby datacenter or CDN.
2. The client opens a TCP (and TLS) connection and sends an HTTP request.
3. A load balancer forwards it to a healthy, non-overloaded web server.
4. The web server reads caches and databases and calls internal services.
5. The HTTP response is returned to the browser.

Reliability: Uptime Institute tiers

Tier	Design requirement	Typical availability	Downtime per year
Tier I	Basic capacity, single path, no redundancy	99.671%	about 28.8 hours
Tier II	Redundant capacity components (e.g. extra UPS)	99.741%	about 22.7 hours
Tier III	Concurrently maintainable, multiple paths	99.982%	about 1.6 hours
Tier IV	Fault tolerant, fully duplicated systems	99.995%	about 26 minutes

- **Common causes of outages:** power failures, cooling problems, network misconfiguration and human error during maintenance.
- **Software reliability:** large services replicate data across several datacenters and fail over automatically, so even the loss of a whole site may be invisible to users.

Performance depends on the whole network path

- **Throughput:** for a path with link rates R1, R2, ..., RN, end-to-end throughput is min{R1, ..., RN}, the rate of the bottleneck link. A 100 Gbps datacenter link cannot help a user on a 100 Mbps access link.
- **Delay:** each hop adds processing, queueing, transmission and propagation delay. Distance matters: light in fiber needs about 5 ms per 1,000 km, so a far-away datacenter always feels slower.
- **Loss:** when packets arrive faster than a link can send them, queues grow; when the buffer is full, packets are dropped.
- **Measurement:** providers track round-trip time (RTT) and tail latency, such as the slowest 1% of requests, not only the average.

Thanks for your listening!

Respondent: 265149-周明宇

Course: Computer Networking

2026/10/26